

# Wenhao He

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## WORK EXPERIENCE

### Full-Stack Software Engineer

New York, NY

Clipp Inc

September 2024 – June 2025

- Evaluated n8n, LangChain, and Botpress for an AI ordering chatbot; led working sessions with senior engineers and drove the team's decision toward Botpress for its conversation structure and faster implementation timeline.
- Optimized GraphQL queries to return nested order variants in single calls, eliminating redundant fetches and speeding up frontend data loading across the ordering flow.
- Built Jest test suites across 8 modules (ordering, products, promotions, vendors, POs, returns, accounting, team) reaching 90%+ branch coverage; caught a resolver data-shape mismatch before sprint freeze that would have silently broken the orders page.

### Full-Stack Software Engineer

New York, NY

CAN International Corp

April 2024 – September 2024

- Owned the React Native mobile app (JavaScript, Firebase, Azure) from kickoff through launch, delivering profile, rewards, payments, and job board features for 300+ members.
- Plugged in Azure Form Recognizer to auto-extract fields from uploaded job description PDFs, cutting a multi-minute manual entry process down to a single upload.

## PROJECTS

### Financial Document Intelligence Agent (Bedrock, LangChain, Chroma, Tavily)

Agentic RAG system for finance and compliance workflows

Apr 2026

- Built a **ReAct agent** using Claude Haiku via AWS Bedrock to route queries between a Chroma vector store and Tavily web search; enforced local-first tool priority via system prompt and a structured Result Summary format that keeps internal corpus answers and external web context strictly separated.
- Engineered a thread-safe Chroma singleton and 21 tests covering ingestion, tool-routing accuracy, and company-mismatch rejection; returns source-attributed answers with filename citations at under **4s** latency and **\$0.004/query** at Haiku token rates.

### ResumeMatch (API Gateway, Lambda, DynamoDB)

Serverless AI Analysis

Dec 2025 – Mar 2026

- Designed and built end-to-end a serverless 4-pass LLM pipeline (Textextract OCR → keyword extraction → match scoring → AI rewrite) processing resumes in **26s** average at **\$0.004/analysis** with 100% reliability and zero throttled requests.
- Added a DynamoDB read-through cache with deterministic SHA-256 hashing, TTL expiry, and gzip compression that drops repeat analyses from 25s to **19ms** with zero Bedrock cost on cache hits.
- Built a Python validation layer with five deterministic checks (umbrella terms, word-boundary matching, abbreviation detection, token overlap, context-aware filtering) that post-processes LLM output and corrects false keyword matches.

### Order Processing System (Spring Boot, PostgreSQL, Kafka, Docker)

Event-driven order backend

Mar 2026 – Apr 2026

- Implemented transactional outbox pattern to eliminate dual-write inconsistency between PostgreSQL and Kafka, ensuring zero dropped events during service failures.
- Built consumer-side idempotency using a processed\_events table with (event\_id, consumer\_group) unique constraint, making all Kafka consumers safe under at-least-once redelivery.
- Enforced order status correctness with an enum-based state machine and @Version optimistic locking, preventing lost updates from concurrent consumer writes.

## TECHNICAL SKILLS

**Languages:** TypeScript, Python, JavaScript, Java, C++, Bash

**Frontend:** React, React Native, Next.js, Angular, Flutter, Tailwind, HTML, CSS

**Backend:** Node.js, Express, GraphQL, Flask, FastAPI, Spring

**Cloud:** Kafka, Elastic, Redis, Kubernetes, AWS (Lambda, API Gateway, DynamoDB, S3, Cognito, CloudFront, Bedrock, Textextract), Azure, Docker

**Data:** MySQL, PostgreSQL, MongoDB, Firebase

**AI/ML:** LangChain, Chroma, TensorFlow, PyTorch, Scikit-learn, RAG, Prompt Engineering

**Testing & Tools:** Jest, Playwright, Cypress, pytest, Git, Jira, Datadog, Figma, Linux

## EDUCATION

University at Buffalo, SUNY

MS in AI (2024) / BS in Computer Science (2022)

AWS Certified Developer – Associate

Issued Feb 2026